Merle Coleman Klein

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Portfolio: https://merlek.github.io/

Software Engineer

Software engineer with 10+ years of professional experience building commerce, subscription, licensing, and authentication systems. Combines Java and Spring backend development with Wolfram Language ERP services and full-stack delivery. Experience owning pricing logic, improving API performance, and integrating payment and identity platforms.

Technical Skills

Languages: Java, Wolfram Language, JavaScript, TypeScript, SQL
Backend: Spring Boot, Spring, REST APIs, MySQL, Tomcat, Gradle
Frontend: Angular, HTML, CSS/SCSS
Integrations: Stripe, Google Play subscriptions, JWT
Engineering: API profiling, batch processing, unit and performance-regression testing, Git, Jira, Codex

Professional Experience

Wolfram Research | Software Engineer / Web Application Developer

2016 - Present

- Assumed primary development ownership of the ERP pricing engine, implementing product and market-specific pricing, subscription-renewal rules, and price-change reporting across currencies.
- Improved site-user management API performance by resolving subscription-site lookup failures, reducing dependencies, and optimizing repeated per-user work.
- Built ERP-backed site-administration APIs and interfaces in Wolfram Account, including concurrent user uploads, progress feedback, and more reliable access management for large institutional sites.
- Integrated Apple Pay, Google Pay, and Alipay through Stripe across online checkout and ERP payment workflows, and added wallet-based billing updates in Wolfram Account.
- Modernized the backend API for online purchasing and subscriptions to Java 21, Spring Boot 3, and Tomcat 10, including OAuth2/OIDC browser authentication and expanded regression and integration coverage; in testing.
- Implemented Android in-app subscriptions across Java services and ERP, supporting Wolfram|Alpha Pro purchases through Google Play and access to subscriptions bought directly from Wolfram.
- Built automated execution-time regression checks into the ERP unit-test framework to help detect performance regressions.
- Implemented a site-license discovery API for AI clients using domain and IP-based availability checks, product access links, and fallback recommendations.

Personal Projects

Health Coach | iPhone Workout Journal | Private external TestFlight beta

https://merlek.github.io/#/projects/health-coach

- Built an iPhone workout journal with Angular, Ionic, TypeScript, and Capacitor/Swift bridges for Apple Health access and protected on-device storage.
- Implemented set logging, history, user-approved progression changes, atomic local saves with recovery, CSV import validation and duplicate prevention, and JSON backup/restore; verified builds and domain/import behavior through automated tests and simulator workflows.

AgentFlow | Project Planning and AI Coordination | In development

- Developed an Angular/TypeScript project workspace with a Node.js server and SQLite persistence for project registration, discovery, planning, and attention tracking.
- Implemented portable project-knowledge exports in JSON and Markdown and bounded public-HTTPS repository acquisition, keeping external project source separate from application-owned state.

Earlier Experience

Wolfram Alpha | Software Engineer

2013 - 2015

- Developed Java subscription and payment features, including promotional codes, student eligibility verification, and product-usage credits.
- Built internal customer-service tools for subscription management, with MySQL database and dynamic web-page development.

Wolfram Research | Software Development Intern

2013

- Developed Java proof-of-concept applications, including an XMPP chatbot for IT and a radar image composer using NOAA forecast data.

Education

University of Illinois, Urbana-Champaign

Bachelor of Science in Computer Science | 2013